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Version Number	Modification Content	Date	Modifier
1.2.3	1. Dispatch Mission Interface adds Composite Robot Type 2. Fields in interface callback except for missionCode, missionStatus are changed to optional	2023.11.22	Chen Wenyue
2.0.0	1. Conveyor AMR Mission add a new field "actionInform" to indicate whether upstream notification is needed after the roller is picked or placed. 2.Carton Transfer Unit Mission add new fields "takeActionInform", "putActionInform" to indicate whether upstream notification is needed after the carton is picked or placed. Pick and place interaction is generally used when connecting to a conveyor line.	20240708	Nandy Li
2.11.1	1. Add the description for the new API: /jobQuery. 2. Update the request parameters and request body examples for forbidden area related APIs and /missionCancel API.	Jan 25, 2025	Hechuan Zhao
2.11.2	1. Add the descriptions for the new API: /queryRobByNodeUuidOrForeignCode,. 2. Update the request parameter and request examples for API: /updateForbiddenAreaStatus.	Feb 14, 2025	Qianwen Zhang
2.11.3	1.Add new API：	20250310	Nandy Li

	/queryContainerAll (Query containers in status of both IN/INSERTED and OUT/REMOVED) 2.Update the API： /containerIn (Appoint container validation code when add container)		
2.11.4	1. Add new API: /robotMoveCarry 2. Add new API: /queryFunctionNode 3. Add one more status (WAITFEEDBACK) for callback(When the workflow step is waiting for feedbacking)	20250414	Hechuan Zhao Nandy Li
2.13.0	1. Add new API: /insertRobot 2. Add new API: /removeRobot 3. Add new API: /chargeRobot	20250414	Hechuan Zhao

-- Mission Related APIs

1 Dispatch Mission (Upstream System -> Kuka System)

- API Info

API Name	**submitMission**
API Description	Dispatch mission to Fleet
API URL	http://[IP:Port]/interfaces/api/amr/submitMission
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

Public Request Parameters

Name	Type	Max Length	Required	Defualt Value	Description	Notes
orgId	String		T		Organization id (or factory code, supplier code)	
requestId	String	32	T		Request Id	Process request idempotency
missionCode	String		T		Mission Code	Mission code should be unique.
missionType	String		T		Mission type: Lift AMR MISSION: RACK_MOVE Conveyor AMR Mission: ROLLER_MOVE Carton Transfer Unit Mission: PICKER_MOVE Forklift AMR Mission: FORKLIFT_MOVE Move Mission: MOVE Composite AMR Mission: ROBOTICS_MOVE	
viewBoardType	String		F		Task board displays task tpes, defined by user	
robotType	String		F		Robot function type: Lift AMR: LIFT Conveyor AMR: ROLLER Carton Transfer Unit: PICKER Forklift AMR: FORKLIFT HCS Box Robot: HCSBOX Composite Robot: COMPOSITE	In principle, it should not be empty; if it is empty, it will be inferred through specific robot or container type codes.
robotModels	List<String>		F		Robot type code	Support specifying multiple robot models, one of which will be selected to execute the mission.
robotIds	List<String>		F		Robot id	Support specifying multiple robots, one of which will be

						selected to execute the mission; if the previous task set lockRobotAfterFinish = true, this field is required.
priority	Integer		F	1	Job Priority, 1-99, the smaller the number, the higher the priority, default is 1	Valid when dispatching batch tasks
containerModelCode	String		F		Container model code	
containerCode	String		F		Container code	
templateCode	String		F		Workflow external code	If the mission is dispatched using the workflow external code, the workflow corresponding to the workflow external code will be created without the need to set missionData.
lockRobotAfterFinish	Boolean		F	false	Whether the robot needs to maintain the task lock status after the workflow ends	Mostly used in scenarios where a new workflow still needs to use the robot for carry after the current workflow ends.
unlockRobotId	String		F		Unlock the current robot from the previous workflow.	
unlockMissionCode	String		F		The mission code to lock the current robot.	Used to verify whether the unlock is correct
idleNode	String		F		After the workflow is completed, the robot will go to the idleNode.	
missionData	List<?>		T		The list of node included in the current workflow, refer to the following section for specific fields. If templateCode is setted, this field can be omitted	

1.1 Lift AMR MISSION(missionType=RACK_MOVE)

- Request Parameters

Name	Type	Max	Required	Default	Description	Notes

		Length		Value		
sequence	Integer		T		The default serial number is 1, which is the starting point of the starting shelf, and subsequent nodes increase in sequence	
position	String		T		Mission target node or area	
type	String		T		Mission node type: Node: NODE_POINT Area: NODE_AREA	
putDown	Boolean		F	false	Decide whether the shelf needs to be lowered at current node	
passStrategy	String		F	AUTO	Resume strategy at current node: Automatically resume: AUTO Manually resume: MANUAL	
waitingMillis	Integer		F	0	The waiting time before automatically triggering the next node task.	Required when passStrategy=true

- Request Body Example


```
{
 "orgId": "UNIVERSAL",
 "requestId": "request202309250001",
 "missionCode": "mission202309250001",
 "missionType": "RACK_MOVE",
 "viewBoardType": "",
 "robotModels": [
  "KMP600I"
 ],
 "robotIds": [
  "14"
 ],
 "robotType": "LIFT",
 "priority": 1,
 "containerModelCode": "10001",
 "containerCode": "1000002",
 "templateCode": "",
 "lockRobotAfterFinish": false,
 "unlockRobotId": "",
 "unlockMissionCode": "",
 "idleNode": "A000000013",
 "missionData": [
  {
   "sequence": 1,
   "position": "M001-A001-45",
   "type": "NODE_POINT",
   "putDown": false,
```

```
      "passStrategy": "AUTO",
      "waitingMillis": 0
    },
    {
      "sequence": 2,
      "position": "M001-A001-40",
      "type": "NODE_POINT",
      "putDown": true,
      "passStrategy": "AUTO",
      "waitingMillis": 0
    }
  ]
}
```


- Response Body Example


```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```


1.2 Conveyor AMR MISSION(missionType=ROLLER_MOVE)

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
sequence	Integer		T		The default serial number is 1, which is the starting point of the starting shelf, and subsequent nodes increase in sequence	
position	String		T		Mission target node or area	
type	String		T		Mission node type: Node: NODE_POINT Area: NODE_AREA	
binCode	String		T		Conatiner code carried by the conveyor amr.	
rollerLevel	Integer		F	1	Level of roller car layers	Required for multi-layer conveyor amr
deviceCode	String		F		Device code	When a point corresponds to multiple devices, this value needs to be passed
actionType	String		F		Actions to be performed at the point Pick up the	Optional when manually pick or place the container

					container: ROLLER_RECEIVE Place the container: ROLLER_SEND	
actionConfirm	Boolean		F	false	Whether confirmation is required before picking/placing	Generally used for interaction with conveyor lines
actionInform	Boolean		F	false	Whether notification is required after picking/placing	Generally used for interaction with conveyor lines

- Request Body Example


```
{
 "orgId": "UNIVERSAL",
 "requestId": "request202309250002",
 "missionCode": "mission202309250002",
 "missionType": "ROLLER_MOVE",
 "viewBoardType": "",
 "robotModels": [
  "KMP600I-1-T2"
 ],
 "robotIds": [
  "15"
 ],
 "robotType": "ROLLER",
 "priority": 1,
 "templateCode": "",
 "lockRobotAfterFinish": false,
 "unlockRobotId": "",
 "unlockMissionCode": "",
 "idleNode": "A000000013",
 "missionData": [
  {
   "sequence": 1,
   "position": "M001-A001-49",
   "type": "NODE_POINT",
   "actionType": "ROLLER_RECEIVE",
   "binCode": "1000002",
   "rollerLevel": 1,
   "deviceCode": "bea8f740-5dcc-11ee-91ba-c7b4e0c01550",
   "actionConfirm": false,
   "actionInform": false
  },
  {
   "sequence": 2,
   "position": "M001-A001-40",
   "type": "NODE_POINT",
   "actionType": "ROLLER_SEND",
   "binCode": "1000002",
   "rollerLevel": 1,
```

"deviceCode": "bea8f740-5dcc-11ee-91ba-c7b4e0c01550",
"actionConfirm": false,
"actionInform": false
}
]
}

Response Body Example

{
"data": null,
"code": "0",
"message": null,
"success": true
}

1.3 Carton Transfer Unit Mission(missionType=PICKER_MOVE)

Name	Type	Max Length	Required	Default Value	Description	Notes
sequence	Integer		T		The default serial number is 1, which is the starting point of the starting shelf, and subsequent nodes increase in sequence	
startPosition	String		T		Start node of the carton	
startSlotCode	String		T		Slot code at the start node	
endPosition	String		T		End node of the carton	
endSlotCode	String		T		Slot code at the end node	
binCode	String		T		Bin code	
takeActionConfirm	Boolean		F	false	Whether to confirm before picking up the container	Generally used for interaction with the conveyor line
takeActionInform	Boolean		F	false	Whether to confirm before picking up the container	Generally used for interaction with the conveyor line
putActionConfirm	Boolean		F	false	Whether to confirm after placing the container	Generally used for interaction with the conveyor line
putActionInform	Boolean		F	false	Whether to confirm after placing the container	Generally used for interaction with the conveyor line

- Request Body Example

{
"orgId": "",
"requestId": "",
"missionCode": "",
"missionType": "",
"viewBoardType": "",

```
  "robotModels": [],
  "robotIds": [],
  "robotType": "",
  "priority": 0,
  "containerModelCode": "",
  "containerCode": "",
  "templateCode": "",
  "lockRobotAfterFinish": true,
  "unlockRobotId": "",
  "unlockMissionCode": "",
  "idleNode": "",
  "missionData": [
   {
    "sequence": 1,
    "binCode": "box-001",
    "startPosition": "",
    "startSlotCode": "",
    "takeActionConfirm": false,
    "takeActionInform": false,
    "endPosition": "",
    "endSlotCode": "",
    "putActionConfirm": false,
    "putActionInform": false
   },
   {
    "sequence": 2,
    "binCode": "box-002",
    "startPosition": "",
    "startSlotCode": "",
    "takeActionConfirm": false,
        "takeActionInform": false,
    "endPosition": "",
    "endSlotCode": "",
    "putActionConfirm": false,
        "putActionInform": false
   },
   {
    "sequence": 3,
    "binCode": "box-003",
    "startPosition": "",
    "startSlotCode": "",
    "takeActionConfirm": false,
        "takeActionInform": false,
    "endPosition": "",
    "endSlotCode": "",
    "putActionConfirm": false,
        "putActionInform": false
   }
  ]
}
```

- Response Body Example


```
{
 "data": null,
 "code": "0",
 "message": null,
 "success": true
}
```


1.4 Forklift AMR Mission (missionType=FORKLIFT_MOVE)

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
sequence	Integer		T		The default serial number is 1, which is the starting point of the starting shelf, and subsequent nodes increase in sequence	
position	String		T		Mission target node or area	
stackNumber	Integer		F	0	Number of layers forked	Indicates the number of layers already stacked at the node
actionConfirm	Boolean		F	false	Whether to confirm before forking or placing the container	

- Request Body Example


```
{
 "orgId": "9001",
 "requestId": "request2023030800001",
 "missionCode": "mission2023030800001",
 "missionType": "FORKLIFT_MOVE",
 "viewBoardType": "W01",
 "robotType": "FORKLIFT",
 "robotModels": [],
 "robotIds": [],
 "priority": 1,
 "containerModelCode": "",
 "containerCode": "",
 "templateCode": "",
 "lockRobotAfterFinish": false,
 "unlockRobotId": "",
 "unlockMissionCode": "",
 "idleNode": "",
 "missionData": [
  {
```

```
    "sequence": 1,
    "position": "",
    "stackNumber": 0,
    "actionConfirm": false
  },
  {
    "sequence": 2,
    "position": "",
    "stackNumber": 0,
    "actionConfirm": false
  }
 ]
}
```


- Response Body Example


```
{
 "data": null,
 "code": "0",
 "message": null,
 "success": true
}
```


1.5 Move Mission(missionType=MOVE)

Name	Type	Max Length	Required	Default Value	Description	Notes
sequence	Integer		T		The default serial number is 1, which is the starting point of the starting shelf, and subsequent nodes increase in sequence	
position	String		T		Mission target node or area	
type	String		T		Mission node type: Node: NODE_POINT Area: NODE_AREA	
passStrategy	String		F	AUTO	Resume strategy at current node: Automatically resume: AUTO Manually resume: MANUAL	
waitingMillis	Integer		F	0	The waiting time before automatically triggering the next node task.	Required when passStrategy=true

- Request Body Example


```
{
 "orgId": "UNIVERSAL",
```

"requestId": "request202309250005",
"missionCode": "mission202309250005",
"missionType": "MOVE",
"viewBoardType": "",
"robotModels": [
"KMP600I"
],
"robotIds": [
"14"
],
"robotType": "LIFT",
"priority": 1,
"templateCode": "",
"lockRobotAfterFinish": false,
"unlockRobotId": "",
"unlockMissionCode": "",
"missionData": [
{
"sequence": 1,
"position": "M001-A001-26",
"type": "NODE_POINT",
"passStrategy": " MANUAL",
"waitingMillis": 0
},
{
"sequence": 2,
"position": "M001-A001-31",
"type": "NODE_POINT",
"passStrategy": " MANUAL",
"waitingMillis": 0
}
]
}

- Response Body Example

{
"data": null,
"code": "0",
"message": null,
"success": true
}

1.6 Composite AMR Mission(missionType=ROBOTICS_MOVE)

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
sequence	Integer		T		The default serial number is 1, which is the starting point of the starting	

					shelf, and subsequent nodes increase in sequence	
position	String		T		Mission target node or area	
type	String		T		Mission node type: Node: NODE_POINT Area: NODE_AREA	
applicationName	String		T		Application name	
params	String		F		Robot Arm Parameters (J SON)	
passStrategy	String		F	AUTO	Resume strategy at current node: Automatically resume: AUTO Manually resume: MANUAL	
waitingMillis	Integer		F	0	The waiting time before automatically triggering the next node task.	Required when passStrategy=true

- Request Body Example


```
{
 "orgId": "UNIVERSAL",
 "requestId": "request202311220001",
 "missionCode": "mission202311220001",
 "missionType": "ROBOTICS_MOVE",
 "viewBoardType": "",
 "robotModels": [
  "KMR-IISY"
 ],
 "robotIds": [
  "15"
 ],
 "robotType": "COMPOSITE",
 "priority": 1,
 "containerModelCode": "10001",
 "containerCode": "1000002",
 "templateCode": "",
 "lockRobotAfterFinish": false,
 "unlockRobotId": "",
 "unlockMissionCode": "",
 "idleNode": "A000000014",
 "missionData": [
```

```
    {
      "sequence": 1,
      "position": "M001-A001-45",
      "type": "NODE_POINT",
      "applicationName ": “app”,
          "params": “{\"key1\": \"value1\",\"key2\": \"value2\"}”,
      "passStrategy": "AUTO",
      "waitingMillis": 0
    },
    {
      "sequence": 2,
      "position": "M001-A001-40",
      "type": "NODE_POINT",
      "applicationName ": “app”,
        "params": “{\"key1\": \"value1\",\"key2\": \"value2\"}”,
      "passStrategy": "AUTO",
      "waitingMillis": 0
    }
  ]
}
```


- Response Body Example


```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```


2 Cancel Mission (Upstream System -> Kuka System）

- API Info

API Name	**missionCancel**
API Description	Cancel mission
API URL	http://[IP:Port]/interfaces/api/amr/missionCancel
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
requestId	String	32	T		Request id, UUID32	

missionCode	String		F		Mission code	One of the three options must be filled in, preferably using the missionCode. The point is used in certain specific scenarios.
containerCode	String		F		Container code	
position	String		F		Node UUID of the mission	
cancelMode	String		T	FORCE	Cancel Mode: FORCE: Force cancel, immediately terminate the current mission. NORMAL: Normal cancel, does not cancel the task the robot is executing, waits for the robot to complete the current task before canceling the workflow. REDIRECT_END: Redirect to the end node of the workflow, does not cancel the task the robot is executing, waits for the robot to complete the current task. If the target node does not require lowering the rack, the vehicle carries the rack to the end node; otherwise, the empty robot moves to the end node. REDIRECT_START: Redirect to the start node of the workflow, does not cancel the task the robot is executing, waits for the robot to complete the current task. If the target node does not require lowering	Currently, only tasks for LIFT robots support all strategies, while other types of robots support FORCE only. However, it still depends on whether the robot supports cancellation.

					the rack, the robot carries the rack back to the start node; otherwise, the empty robot moves back to the start node.	
reason	String		F		Reason to cancel	

- Request Body Example

```
{
 "requestId": "request202309250006",
 "missionCode": "mission202309250004",
 "containerCode": "",
 "position": "",
 "cancelMode": "FORCE",
 "reason": ""
}
```

- Response Body Example

```
{
 "data": null,
 "code": "0",
 "message": null,
 "success": true
}
```

3 Resume Workflow (Upstream System -> Kuka System）

- API Info

API Name	**operationFeedback**
API Description	Feedback on task completion information from the upstream after the node task is completed
API URL	http://[IP:Port]/interfaces/api/amr/operationFeedback
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Description	Notes
requestId	String	32	T	Request id, UUID32	
missionCode	String		F	Mission code of the running workflow	Three choose one
containerCode	String		F	Container code of the running workflow	
position	String		F	The current node of the workflow	

- Request Body Example

```
{
    "requestId": "request202309250007",
    "containerCode": "",
    "missionCode": "mission202309250005",
    "position": ""
}
```

- Response Body Example

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

4 Query Jobs (Upstream System -> Kuka System)

- API Info

API Name	jobQuery
API Description	Query jobs in the job board
API URL	http://[IP:Port]/interfaces/api/amr/jobQuery
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
workflowId	Long		F		Workflow id	
containerCode	String		F		Conatiner code	
jobCode	String		F		Job code	Same with the mission code
status	String		F		Job status: Created: 10 Executing: 20 Waiting: 25 Cancelling: 28 Complete: 30 Cancelled: 31 Manual complete: 35 Warning: 50 Startup error: 60	
robotId	String		F		Robot id	
targetCellCode	String		F		Target node code of the running task in the	

					workflow	
workflowName	String		F		Name of the workflow configuration	
workflowCode	String		F		Workflow code of the workflow configuration	
maps	List<String>		F		Map code where the workflow is running	
createUsername	String		F		Operator	
sourceValue	Integer		F		Source: Interface: 2 PDA: 3 Device: 4 MLS: 5 Fleet application: 6 Workflow event: 7	
limit	Integer		F	10	Number of jobs in the response, default 10.	

- Response Parameters

Name	Type	Description	Notes
jobCode	String	Job code	Same with the mission code
workflowId	Long	Workflow id	
containerCode	String	Conatiner code	
robotId	String	Robot id	
status	Integer	Job status: Created: 10 Executing: 20 Waiting: 25 Cancelling: 28 Complete: 30 Cancelled: 31 Manual complete: 35 Warning: 50 Startup error: 60	
workflowName	String	Name of the workflow configuration	Name could be empty when the workflow is generated by upstream system
workflowCode	String	Workflow code of the workflow configuration	Workflow code could be empty when the workflow is generated by upstream system
workflowPriority	Integer	Workflow priority	
mapCode	String	Map code where	
targetCellCode	String	Target node code of the running task in the workflow	
beginCellCode	String	Begin node code of the running task in the workflow	
targetCellCodeForeign	String	Foreign code of target	

		node code of the running task in the workflow	
beginCellCodeForeign	String	Foreign code of begin node code of the running task in the workflow	
finalNodeCode	String	End node of the workflow	Area code for area node and node code for the map node
warnFlag	Integer	Warn flag: Normal: 0 Warning: 1	
warnCode	String	Warn code	
completeTime	String	Workflow complete time (yyyy-MM-dd HH:mm:ss)	
spendTime	Integer	Workflow spend time (second)	
createUsername	String	Operator	
createTime	String	Workflow create time (yyyy-MM-dd HH:mm:ss)	
source	String	Source: INTERFACE, PDA, DEVICE, MLS, SELF, EVENT	
materialsInfo	String	Materials info	Require when users need material management of containers

- Request Body Example(for reference only, fill in the parameters as needed, all fields are optional)

```
{
    "containerCode": "C001",
    "createUsername": "admin",
    "jobCode": "T000096284",
    "limit": 10,
    "maps": ["TEST"],
    "robotId": "1",
    "sourceValue": 6,
    "status": 20,
    "targetCellCode": "TEST-1-90",
    "workflowCode": " W000000587",
    "workflowId": 100218,
    "workflowName": "Carry01"
}
```

- Response Body Example

```
{
 "data": [
  {
   "jobCode": "T000096284",
   "workflowId": 100218,
   "containerCode": "C001",
   "robotId":"1",
   "status": 20,
   "workflowName": "Carry01",
   "workflowCode": " W000000587",
```

"workflowPriority": 1,
"mapCode": "TEST",
"targetCellCode": "TEST-1-90",
"beginCellCode": "TEST-1-80",
"targetCellCodeForeign": "DROPPOINT",
"beginCellCodeForeign": "PICKPOINT",
"finalNodeCode": "TEST-1-90",
"warnFlag": 0,
"warnCode": null
"completeTime": null,
"spendTime": null,
"createUsername": "admin",
"createTime": "2025-01-10 16:01:42",
"source": "SELF",
"materialsInfo": "-"
}],
"code": "0",
"message": null,
"success": true
}

-- Container Related APIs

5 Insert Container (Upstream System -> Kuka System）

- API Info

API Name	containerIn
API Description	Insert the container
API URL	http://[IP:Port]/interfaces/api/amr/containerIn
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
requestId	String	32	T		Request id, UUID32	
containerType	String		F		Container type: RACK, BIN	
containerModelCode	String		F		Container model code	Required when isNew = true

containerCode	String		T		Container code	
enterOrientation	String	32	F		Container orientation after insert	Special requirements for the orientation when inserting the container.
position	String		T		Node code where inserting the container	
isNew	Boolean		F	false	Automaticly create the container before inserting	
containerValidationCode	String		F		Appointed validation code when creating the container	When isNew=true, appointed validation code
withDefaultValidationCode	Boolean		F		Using default validation code when creating the container(the same with containerCode)	When isNew=true, set the container validation code same with containerCode

- Request Body Example(Insert)

```
{
 "requestId": "request202309250008",
 "containerCode": "10",
 "position": "M001-A001-31",
 "containerModelCode": "",
 "enterOrientation": "",
 "isNew": false
}
```

- Request Body Example(Add and Insert with Appointed validation Code)

```
{
 "requestId": "request202309250008",
 "containerCode": "10",
 "position": "M001-A001-31",
 "containerModelCode": "1",
 "enterOrientation": "0",
 "isNew": true,
  "containerValidationCode": "ID:10",
  "withDefaultValidationCode": false
}
```

- Request Body Example(Add and Insert with Default validation Code)

```
{
 "requestId": "request202309250008",
 "containerCode": "10",
 "position": "M001-A001-31",
 "containerModelCode": "1",
 "enterOrientation": "0",
 "isNew": true,
```

"containerValidationCode": "",
"withDefaultValidationCode": true
}

- Response Body Example

{
"data": null,
"code": "0",
"message": null,
"success": true
}

6 Remove Container (Upstream System -> Kuka System）

- API Info

API Name	**containerOut**
API Description	Remove the container
API URL	http://[IP:Port]/interfaces/api/amr/containerOut
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
requestId	String	32	T		Request id, UUID32	
containerType	String		F	RACK	Container type: RACK	
containerCode	String		T		Container code	
position	String		F		Node code of the container	
isDelete	Boolean		F	false	Delete container after removing the container	

- Request Body Example

{
"requestId": "request202309250009",
"containerType": "",
"containerCode": "10",
"position": "M001-A001-31",
"isDelete": false
}

- Response Body Example

{
"data": null,
"code": "0",
"message": null,

"success": true
}

7 Update Container (Supported: updates position and empty full status）(Upstream System -> Kuka System）

- API Info

API Name	**updateContainer**
API Description	Update container
API URL	http://[IP:Port]/interfaces/api/amr/updateContainer
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
requestId	String	32	T		Request id, UUID32	
containerType	String		F		Container type: RACK	
containerCode	String		T		Container code	
originPosition	String		F		Origin node code of the container	
targetPosition	String		T		Target node code of the container	
emptyStatus	String		F		Container empty/full status: EMPTY, FULL	
reason	String		F		Reason to update	

- Request Body Example

{
"requestId": "request202309250010",
"containerType": "BUCKET",
"containerCode": "10",
"originPosition": "M001-A001-31",
"targetPosition": "M001-A001-30",
"emptyStatus": "EMPTY",
"reason": ""
}

- Response Body Example

{
"data": null,
"code": "0",
"message": null,
"success": true

}

8 Mission Status Callback (Upstream System -> Kuka System）

- API Info

API Name	**missionStateCallback**
API Description	Mission status callback
API URL	http://[IP:Port]/interfaces/api/amr/missionStateCallback
HTTP Method	POST

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
missionCode	String	32	T		Mission code	Same with the jobcode of the workflow
viewBoardType	String		F		View board type	
containerCode	String		F		Conatiner code	
currentPosition	String		F		Current node code of the mission	
slotCode	String		F		Current slot code	
robotId	String		F		Robot which is executing the mission	
missionStatus	String		T		Mission Status: MOVE_BEGIN, ARRIVED, UP_CONTAINER, DOWN_CONTAINER, ROLLER_RECEIVE, ROLLER_SEND, PICKER_RECEIVE, PICKER_SEND, FORK_UP, FORK_DOWN, WAITFEEDBACK, COMPLETED, CANCELED	
message	String		F		Description message	
missionData			F		Custom information object that needs to be reported	

- Request Body Example

```
{
 "missionCode": "mission202309250005",
 "viewBoardType": "",
 "slotCode": "",
 "robotId": "14",
 "containerCode": "1000002",
 "currentPosition": "M001-A001-31",
 "missionStatus": "MOVE_BEGIN",
 "message": "",
 "missionData": {}
```

}

9 Query Container Model Codes (Upstream System -> Kuka System）

- API Info

API Name	queryAllContainerModelCode
API Description	Query all container models
API URL	http://[IP:Port]/interfaces/api/amr/queryAllContainerModelCode
HTTP Method	GET

- No Request example
- Response Body Example

{
 "code": "0",
 "message": null,
 "success": true,
 "data": [
  "10001"
 ]
}

10 Query Area Codes By Conatiner Model (Upstream System -> Kuka System）

- API Info

API Name	queryAreaCodeForContainerModel
API Description	Query areas that supports the current container model
API URL	http://[IP:Port]/interfaces/api/amr/queryAreaCodeForContainerModel
HTTP Method	GET

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
containerModelCode	String		T		Container model code	
noContainerFirst	Boolean		F	false	是否区域内无容器优先： ture 是 false 否	

- Request URL Example

http://[IP:Port]/interfaces/api/amr/queryAreaCodeForContainerModel?containerModelCode=10001&noContainerFirst=false

- Response Body Example

{
 "code": "0",
 "message": "",

```
  "success": true,
  "data": [
    "areaCode1",
    "areaCode2",
    "areaCode3"
  ]
}
```


11-1 Query Containers(Only including containers with IN/INSERTED status) (Upstream System -> Kuka System）

- API Info

API Name	containerQuery
API Description	Query containers
API URL	http://[IP:Port]/interfaces/api/amr/containerQuery
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
containerCode	String		F		Container code	
nodeCode	String		F		Node code	
areaCode	String		F		Area code	
containerModelCode	String		F		Container model code	
emptyFullStatus	Integer		F	2	Container empty/full status: EMPTY: 0 FULL: 1 ALL: 2	The empty/full status will only take effect when the area code is not empty.

- Response Parameters

Name	Type	Description	Notes
containerCode	String	Container code	
nodeCode	String	Node code	
inMapStatus	Integer	In map status: In map: 1 Off map: 0	
containerModelCode	String	Container model code	
emptyFullStatus	String	Container empty/full status: EMPTY: 0 FULL: 1	

isCarry	Integer	Lift status: Drop: 0 Lift: 1	
orientation	String	Container orientation	
containerCheckCode	String	Container check code	
mapCode	String	Map code	
districtCode	String	Map district code	

- Request Body Example


```
{
 "nodeCode": "M001-A001-30",
 "containerModelCode": "10001",
 "containerCode": "10",
 "areaCode": "A000000014",
 "emptyFullStatus": 2
}
```


- Response Body Example


```
{
 "data": [
  {
   "containerCode": "10",
   "nodeCode": "",
   "orientation": "0.0",
   "containerModelCode": "10001",
   "emptyFullStatus": 0,
   "inMapStatus": 1,
   "isCarry": 0,
   "containerCheckCode": "10",
   "mapCode": "ABC",
   "districtCode": "1"
  }
 ],
 "code": "0",
 "message": null,
 "success": true
}
```


11-2 Query Containers(Both IN/INSERTED and OUT/REMOVED status) (Upstream System -> Kuka System）

- API Info

API Name	containerQuery
API Description	Query containers
API URL	http://[IP:Port]/interfaces/api/amr/containerQuery
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
containerCode	String		F		Container code	
nodeCode	String		F		Node code	
areaCode	String		F		Area code	
mapCode	String		F		Map code	
districtCode	String		F		District code	
containerModelCode	String		F		Container model code	
emptyFullStatus	Integer		F		Container empty/full status: EMPTY: 0 FULL: 1	
inMapStatus	Integer		F		IN/OUT map status IN: 1 OUT:0	

- Response Parameters

Name	Type	Description	Notes
containerCode	String	Container code	
nodeCode	String	Node code	
inMapStatus	Integer	In map status: In map: 1 Off map: 0	
containerModelCode	String	Container model code	
emptyFullStatus	String	Container empty/full status: EMPTY: 0 FULL: 1	
isCarry	Integer	Lift status: Drop: 0 Lift: 1	
orientation	String	Container orientation	
containerCheckCode	String	Container check code	
mapCode	String	Map code	
districtCode	String	Map district code	

- Request Body Example(Query all container in map ABC)

```
{
  "mapCode": "ABC"
}
```

- Response Body Example

```
{
  "data": [
    {
      "containerCode": "10",
      "nodeCode": "1-1",
```

```
    "orientation": "0.0",
    "containerModelCode": "10001",
    "emptyFullStatus": 0,
    "inMapStatus": 1,
    "isCarry": 0,
    "containerCheckCode": "10",
    "mapCode": "ABC",
    "districtCode": "1"
  },
  {
    "containerCode": "20",
    "nodeCode": "1-1",
    "orientation": "0.0",
    "containerModelCode": "10001",
    "emptyFullStatus": 0,
    "inMapStatus": 0,
    "isCarry": 0,
    "containerCheckCode": "",
    "mapCode": "ABC",
    "districtCode": "1"
  }
 ],
 "code": "0",
 "message": null,
 "success": true
}
```


-- Robot Module

12 Query Robot by Node UUID or Node Code (Upstream System -> Kuka System)

- API Info

API Name	**queryRobByNodeUuidOrForeignCode**
API Description	Query robot by node UUID or node foreign code
API URL	http://[IP:Port]/interfaces/api/amr/queryRobByNodeUuidOrForeignCode
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
nodeCode	String		T		Node UUID or node foreign code	If querying multiple robots, the user can pass multiple node parameters separated by commas.

- Response Parameters

Name	Type	Description	Notes
robotId	String	Robot id	
robotType	String	Robot type code	
mapCode	String	Map code	
floorNumber	String	Floor number	
buildingCode	String	Building code	
containerCode	String	Container code which is carried by the robot	
status	Integer	Departure: 1 Offline: 2 Idle: 3 Executing: 4 Charging: 5 Updating:6 Abnormal: 7	
occupyStatus	Integer	Occupy status: Occupied: 1 Idle: 0	
batteryLevel	Double	Battery level (%)	
nodeCode	String	Node code	
X	String	Robot current location Y (millimeters)	
Y	String	Robot current location X (millimeters)	
robotOrientation	String	Robot orientation	
missionCode	String	Mission code	Mission code dispatched by the upstream system
liftStatus	Integer	Robot lift status: Lift: 1	
reliability	Integer	Location reliability: Unreliable: 0 Reliable: 1	
runTime	String	Run time	
karOsVersion	String	KarOs version	
mileage	String	Running mileage	
leftMotorTemperature	String	Left motor temperature	
rightMotorTemperature	String	Right motor temperature	
liftMotorTemperature	String	Lift motor tempreature	
rotateMotorTemperature	String	Rotate motor temperature	
rotateTimes	Integer	Rotate times	
liftTimes	Integer	Lift times	
nodeForeignCode	String	Foreign code of the node	When querying robots using the node

		where the robot locates	foreign code, this field will have a value.

- Request Example:
  1) Example 1

  ```
  {
      "nodeCode": "startNode1, startNode2"
  }
  ```


  2) Example 2

  ```
  {
      "nodeCode": "test01-5-91"
  }
  ```

- Response Body Example
  1) Example 1

  ```
  {
   "data": [
    {
     "robotId": "925",
     "robotType": "KMP600I",
     "mapCode": "test01",
     "floorNumber": "5",
     "buildingCode": "test",
     "containerCode": "",
     "status": 3,
     "occupyStatus": 0,
     "batteryLevel": 1,
     "nodeCode": "test01-5-64",
     "x": "3000.0",
     "y": "15000.0",
     "robotOrientation": "90.0",
     "missionCode": null,
     "liftStatus": 0,
     "reliability ": 1,
     "runTime ": "6745",
     "karOsVersion ": "",
     "mileage ": "0.0",
     "leftMotorTemperature ": "0.0",
     "rightMotorTemperature ": "0.0",
     "liftMotorTemperature ": "0.0",
     "rotateMotorTemperature ": "0.0",
     "rotateTimes ": 0,
     "liftTimes ": 0,
     "nodeForeignCode": " startNode1"
    },
    {
     "robotId": "80211",
     "robotType": "KMP600I",
     "mapCode": "test01",
     "floorNumber": "5",
     "buildingCode": "test",
  ```


```
    "containerCode": "",
    "status": 3,
    "occupyStatus": 0,
    "batteryLevel": 1,
    "nodeCode": "test01-5-74",
    "x": "1200.0",
    "y": "2400.0",
    "robotOrientation": "0.0",
    "missionCode": null,
    "liftStatus": 0,
    "reliability ": 1,
    "runTime ": "null",
    "karOsVersion ": "",
    "mileage ": "0.0",
    "leftMotorTemperature ": "0.0",
    "rightMotorTemperature ": "0.0",
    "liftMotorTemperature ": "0.0",
    "rotateMotorTemperature ": "0.0",
    "rotateTimes ": 0,
    "liftTimes ": 0,
    "nodeForeignCode": " startNode2"
  }
 ],
 "code": "0",
 "message": null,
 "success": true
}
```


2) Example 2


```
{
 "data": [
  {
    "robotId": "2041",
    "robotType": "KMP600I",
    "mapCode": "test01",
    "floorNumber": "5",
    "buildingCode": "test",
    "containerCode": "",
    "status": 3,
    "occupyStatus": 0,
    "batteryLevel": 1,
    "nodeCode": "test01-5-91",
    "x": "13000.0",
    "y": "30000.0",
    "robotOrientation": "-90.0",
    "missionCode": null,
    "liftStatus": 0,
    "reliability ": 1,
    "runTime ": "6745",
    "karOsVersion ": "",
    "mileage ": "0.0",
```

```
    "leftMotorTemperature ": "0.0",
    "rightMotorTemperature ": "0.0",
    "liftMotorTemperature ": "0.0",
    "rotateMotorTemperature ": "0.0",
    "rotateTimes ": 0,
    "liftTimes ": 0,
    "nodeForeignCode": ""
  }
 ],
 "code": "0",
 "message": null,
 "success": true
}
```


13 Query Robots (Upstream System -> Kuka System)

- API Info

API Name	robotQuery
API Description	Query robot (default to search all)
API URL	http://[IP:Port]/interfaces/api/amr/robotQuery
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
robotId	String		F		Robot id	If all parameters are empty, then query all by default. **Note:** The request parameters mapCode and floorNumber must be passed together.
robotType	String		F		Robot type code	
mapCode	String		F		Map code (If map is disabled, result is empty)	
floorNumber	String		F		Floor number (If map is disabled, result is empty)	

- Response Parameters

Name	Type	Description	Notes
robotId	String	Robot id	
robotType	String	Robot type code	

mapCode	String	Map code	
floorNumber	String	Floor number	
buildingCode	String	Building code	
containerCode	String	Container code which is carried by the robot	
status	Integer	Departure: 1 Offline: 2 Idle: 3 Executing: 4 Charging: 5 Updating:6 Abnormal: 7	
occupyStatus	Integer	Occupy status: Occupied: 1 Idle: 0	
batteryLevel	Double	Battery level (%)	
nodeCode	String	Node code	
X	String	Robot current location Y (millimeters)	
Y	String	Robot current location X (millimeters)	
robotOrientation	String	Robot orientation	
missionCode	String	Mission code	Mission code dispatched by the upstream system
liftStatus	Integer	Robot lift status: Lift: 1	
reliability	Integer	Location reliability: Unreliable: 0 Reliable: 1	
runTime	String	Run time	
karOsVersion	String	KarOs version	
mileage	String	Running mileage	
leftMotorTemperature	String	Left motor temperature	
rightMotorTemperature	String	Right motor temperature	
liftMotorTemperature	String	Lift motor tempreature	
rotateMotorTemperature	String	Rotate motor temperature	
rotateTimes	Integer	Rotate times	
liftTimes	Integer	Lift times	

- Request Body Example

```
{
    "floorNumber": "A001",
    "mapCode": "M001",
    "robotId": "13",
    "robotType": "KMP600I"
}
```

- Response Body Example

```
{
  "data": [
    {
      "robotId": "13",
      "robotType": "KMP600I",
```

"mapCode": "M001",
"floorNumber": "A001",
"buildingCode": "W001",
"containerCode": "",
"status": 3,
"occupyStatus": 0,
"batteryLevel": 1,
"nodeCode": "46",
"x": "3000.0",
"y": "15000.0",
"robotOrientation": "90.0",
"missionCode": null,
"liftStatus": 0,
"reliability ": 1,
"runTime ": "6745",
"karOsVersion ": "",
"mileage ": "0.0",
"leftMotorTemperature ": "0.0",
"rightMotorTemperature ": "0.0",
"liftMotorTemperature ": "0.0",
"rotateMotorTemperature ": "0.0",
"rotateTimes ": 0,
"liftTimes ": 0
}
],
"code": "0",
"message": null,
"success": true
}

14 Dispatch Move Carry Task (Upstream System -> Kuka System)

- API Info

API Name	**robotMoveCarry**
API Description	Dispatch a move carry task to the robot. The function is the same as which in the Map Monitor, directly dispatched to the Mission Manager, only displayed in the Mission List, and not visible on the Job Board.
API URL	http://[IP:Port]/interfaces/api/amr/robotMoveCarry
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
robotId	String		T		Robot Id	
containerCode	String		T		Container code	
targetNodeCode	String		T		Target node code	
missionCode	String		F		Mission code which dispatched to the Mission Manager.	Automatically generate the mission code, if the parameter is not passed.

- Request Body Example

```
{
    "robotId": "10001",
    "containerCode": "C0002",
    "targetNodeCode": "TEST-1-45"
}
```

- Response Body Example

```
{
 "data": "I-CARRY-1744609148201",
 "code": "0",
 "message": "OK",
 "success": true
}
```

15 Dispatch Charge Robot Task (Upstream System -> Kuka System)

- API Info

API Name	**chargeRobot**
API Description	Dispatch charge task to the robot. The function is the same as which in the Map Monitor, directly dispatched to the Mission Manager, only displayed in the Mission List, and not visible on the Job Board.
API URL	http://[IP:Port]/interfaces/api/amr/chargeRobot
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
robotId	String		T		Robot Id	

necessary	Integer		T		Whether to force generating the charging task.	1: Always generate a task and queue it; 0: Charge when below lowestLevel, task can be directly completed if no resources available
targetNodeCode	String		T		Target node code	
targetLevel	Integer		T		Target battery level	
lowestLevel	Integer		F		Lowest battery level to trigger charge	Required when necessary = 0
missionCode	String		F		Mission code which dispatched to the Mission Manager.	Automatically generate the mission code, if the parameter is not passed.

- Request Body Example

```
{
    "lowestLevel": 5,
    "necessary": 0,
    "robotId": "2",
    "targetLevel": 90
}
```

- Response Body Example

```
{
 "data": null,
 "code": "0",
 "message": "OK",
 "success": true
}
```

16 Insert The Robot Into The Map (Upstream System -> Kuka System)

- API Info

API Name	**insertRobot**
API Description	Insert the robot into the map. Only support real AMR currently (distinguished from simulated AMR).
API URL	http://[IP:Port]/interfaces/api/amr/insertRobot
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
robotId	String		T		Robot id	
cellCode	String		T		Node code to insert the robot	
synchroContainer	Integer		T		Whether to insert with the container.	0: No 1: Yes

- Request Body Example

```
{
    "cellCode": "TEST-1-69",
    "robotId": "2",
    "synchroContainer": 1
}
```

- Response Body Example

```
{
 "data": null,
 "code": "0",
 "message": "OK",
 "success": true
}
```

17 Remove The Robot From The Map (Upstream System -> Kuka System)

- API Info

API Name	removeRobot
API Description	Remove the robot from the map.
API URL	http://[IP:Port]/interfaces/api/amr/removeRobot
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
robotId	String		T		Robot Id	
withContainer	Integer		F	1	Whether to remove with	0: No

					the container.	1: Yes

- Request Body Example

```
{
    "robotId": "2",
    "withContainer": 1
}
```

- Response Body Example

```
{
 "data": null,
 "code": "0",
 "message": "OK",
 "success": true
}
```

-- AMR Map Node and Area Related Module

18 Query All WCS Area Information (Upstream System -> Kuka System)

- API Info

API Name	**areaQuery**
API Description	查询所有 WCS 区域信息 Query All WCS Area Information
API URL	http://[IP:Port]/interfaces/api/amr/areaQuery
HTTP Method	GET

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

None

- Response Parameters

Name	Type	Description	Notes
areaName	String	Area Name	
areaCode	String	Area Code	
areaType	Integer	Area Type 1: container storage area，3:temporary storage area, 4:buffer area, 5:dense storage area, 6:exclusive area, 7: fork area	

- Request Body Example
- Response Body Example


```
{
 "data": [
  {
   "areaName": "container storage area",
   "areaType": 1,
   "areaCode": "1-song-1732701220548"
  },
  {
   "areaName": "temporary storage area",
   "areaType": 3,
   "areaCode": "1-song-1732701269477"
  },
  {
   "areaName": "buffer area",
   "areaType": 4,
   "areaCode": "1-song-1732701316971"
  },
  {
   "areaName": "dense storage area",
   "areaType": 5,
   "areaCode": "1-song-1732701347592"
  },
  {
   "areaName": "exclusive area",
   "areaType": 6,
   "areaCode": "1-song-1732701422092"
  },
  {
   "areaName": "fork area",
   "areaType": 7,
   "areaCode": "1-song-1732701459113"
  }
 ],
 "code": "0",
 "message": null,
 "success": true
}
```


19 Query Nodes of the Area (Upstream System -> Kuka System)

- API Info

API Name	**areaNodesQuery**
API Description	Query nodes of the area
API URL	http://[IP:Port]/interfaces/api/amr/areaNodesQuery

HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
areaCodes	List<String>		T		Area codes	

- Response Parameters

Name	Type	Description	Notes
areaCode	String	Area code	
nodeList	List<String>	The set of external codes for the nodes. If all points in the area have no external codes, this field may be empty.	

- Request URL Example

1） Example 1

```
{
    "areaCodes": ["A000000013","A000000014"]
}
```

2） Example 2

```
{
    "areaCodes": ["1-song-1732612877805"]
}
```

- Response Body Example

1）Example 1

```
{
 "data": [
  {
   "areaCode": "A000000014",
   "nodeList": [
    "27"
   ]
  },
  {
   "areaCode": "A000000013",
   "nodeList": [
    "46",
    "50"
   ]
  }
```

```
  ],
  "code": "0",
  "message": null,
  "success": true
}
```


3） Example 2，all nodes within the area have no external codes


```
{
  "data": [
    {
      "areaCode": "1-song-1732612877805",
      "nodeList": [
        ""
      ]
    }
  ],
  "code": "0",
  "message": null,
  "success": true
}
```


20 Query Area by Map Node (Upstream System -> Kuka System)

- API Info

API Name	**queryWCSAreaByMapNode**
API Description	Query area by map node
API URL	http://[IP:Port]/interfaces/api/amr/queryWCSAreaByMapNode
HTTP Method	GET

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
nodeUuid	String		T		Node UUID	

- Response Parameters

Name	Type	Description	Notes
areaCode	String	Area code	
containerModelCode	String	Model code of containers	Not required

		that can be placed in the area	

- Request URL Example
  http://[IP:Port]/interfaces/api/amr/queryWCSAreaByMapNode?nodeUuid= 1-song-44

- Response Body Example
  1) Query the area to which the node belongs, and the area is not configured with a container model code that can be placed


```
{
 "data": {
  "areaCode": "1-song-1732612831825",
  "containerModelCode": ""
 },
 "code": "0",
 "message": null,
 "success": true
}
```


2. Query the area to which the node belongs, and the area is not configured with a container model code that can be placed.

2)


```
{
 "data": {
  "areaCode": "1-song-1732612877805",
  "containerModelCode": "600i"
 },
 "code": "0",
 "message": null,
 "success": true
}
```


3）Can not find the area to which the point belongs


```
{
 "data": null,
 "code": "100001",
 "message": "No such node in the graph.[7788]",
 "success": false
}
```


21 Query All Forbidden Areas (Upstream System -> Kuka System)

- API Info

API Name	queryAllForbiddenAreas

API Description	Query all forbidden areas
API URL	http://[IP:Port]/interfaces/api/amr/queryAllForbiddenAreas
HTTP Method	GET

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

None

- Response Parameters

Name	Type	Description	Notes
forbiddenAreaId	Integer	Forbidden area ID	
mapCode	String	Map code where the area locates	
floorNumber	String	District code where the area locates	
description	String	Area description	
forbiddenAreaStrategy	Integer	Area strategy:0-default;1-allow AMR parking	
status	Integer	Status: 0-disabled; 1-enabled; 2-effective; 3-ineffective	
enableType	Integer	Effective strategy: 0-immediately; 1-effective according to the given time period	

- Request Body Example

None

- Response Body Example


```
{
 "data": [
  {
   "forbiddenAreaId": 3,
   "mapCode": "1",
   "floorNumber": "song",
   "description": "",
   "forbiddenAreaStrategy": 0,
   "status": 1,
   "enableType": 0
  },
  {
   "forbiddenAreaId": 4,
   "mapCode": "1",
```

"floorNumber": "1",
"description": "",
"forbiddenAreaStrategy": 0,
"status": 1,
"enableType": 0
},
{
"forbiddenAreaId": 2,
"mapCode": "1",
"floorNumber": "song",
"description": "forbidden area1 of map song",
"forbiddenAreaStrategy": 0,
"status": 1,
"enableType": 0
}
],
"code": "0",
"message": null,
"success": true
}],
"code": "0",
"message": null,
"success": true
}

22 Query One Forbidden Area (Upstream System -> Kuka System)

- API Info

API Name	queryOneForbiddenArea
API Description	Query one forbidden area
API URL	http://[IP:Port]/interfaces/api/amr/queryOneForbiddenArea
HTTP Method	GET

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
forbiddenAreaId	Integer		T		Forbidden area ID	

- Response Parameters

Name	Type	Description	Notes

forbiddenAreaId	Integer	Forbidden area ID	
mapCode	String	Map cpde where the area locates	
floorNumber	String	District code where the area locates	
description	String	Area description	Not required
forbiddenAreaStrategy	Integer	Area strategy:0-default;1-allow AMR parking	
status	Integer	Status: 0-disabled; 1-enabled; 2-effective; 3-ineffective	
enableType	Integer	Effective strategy: 0-immediately; 1-effective according to the given time period	

- Request URL Example
  http://[IP:Port]/interfaces/api/amr/queryOneForbiddenArea?forbiddenAreaId=2

- Response Body Example

Example 1


```
{
 "data": {
  "forbiddenAreaId": 2,
  "mapCode": "1",
  "floorNumber": "song",
  "description": "forbidden area1 of map song",
  "forbiddenAreaStrategy": 0,
  "status": 1,
  "enableType": 0
 },
 "code": "0",
 "message": null,
 "success": true
}
```


23 Update Forbidden Area Status (Upstream System -> Kuka System)

- API Info

API Name	**updateForbiddenAreaStatus**
API Description	Update forbidden area status
API URL	http://[IP:Port]/interfaces/api/amr/updateForbiddenAreaStatus
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
forbiddenAreaId	Integer		T	0	Forbidden area ID	Only one of forbiddenAreaId or forbiddenAreaCode can be selected as a parameter in the request body at a time.
forbiddenAreaCode	String		T		Forbidden area code	
status	String		T		Forbidden area status: 0-disabled; 1-enabled	

- Response Parameters

None

- Request Body Examples

1. Example 1

```
{
  "forbiddenAreaId": 2,
  "status": "0"
}
```

2. Example 2

```
{
    "forbiddenAreaCode": "map1-01-1739427001143",
   "status": "0"
}
```

- Response Body Example

1) Modify status successfully

```
{
  "data": null,
  "code": "0",
  "message": null,
  "success": true
}
```

2) Fail to modify status

Error reason: The status of the forbidden area is already 0 (disabled). If you still attempt to change its status to 0, you will receive the following error. If the status is already 1 (enabled) and you attempt to change its status to 1 again, you will receive a similar error.

```
{
  "data": null,
  "code": "100001",
  "message": "RCS returns error[[D]com.kuka.rcs.bd.common.grpc.util.CommonGrpcErrorCode.16[can not update [ MAP_ZONE_STATUS_DISABLE ]]]",
  "success": false
}
```

3) an error in passing the parameters

Error reason: Only one of forbiddenAreaId or forbiddenAreaCode can be selected as a parameter in the request body at a time.


```
{
  "data": null,
  "code": "100001",
  "message": "only one of area ID or area code can be filled in",
  "success": false
}
```


24 Query Function Node (Upstream System -> Kuka System)

- API Info

API Name	**queryFunctionNode**
API Description	Query function node
API URL	http://[IP:Port]/interfaces/api/amr/queryFunctionNode
HTTP Method	POST

- Headers

Param Name	Param Value	Required	Example	Notes
Content-Type	application/json	T		

- Request Parameters

Name	Type	Max Length	Required	Default Value	Description	Notes
functionType	Integer		T		Node function type	1: Workstation 2: Shelf point 3: Rest point 4: Charging entry point 5: Interaction point 6: Avoidance point 7: Rotation point 8: Queuing point 9: Check point 10: Charging point 11: Elevator point 12: Box shelf point 13: Pipeline point 15: Voice broadcast point 16: Mobile box shelf point 17: Pipeline relative point 18: Wireless charging point 19: Connect point

						20: CTU loader node point
mapCode	String		T		Map code	
floorNumber	String		T		Floor number	
robotTypeClass	String		F		Robot type class	
containerModel	String		F		Container model code	

- Response Parameters

Name	Type	Description	Notes
nodeCode	String	Node code	
externalCode	String	Node external code	
functionType	Integer	Node function type	
containerModelCode	List<String>	Node supported container model code	
robotTypeCode	List<String>	Node supported robot type code	

- Request Body Examples

```
{
    "floorNumber": "1",
    "functionType": 1,
    "mapCode": "TEST"
}
```

- Response Body Example

```
{
 "data": [
  {
   "nodeCode": "TEST-1-53",
   "externalCode": "",
   "functionType": 1,
   "containerModelCode": [
    "model1"
   ],
   "robotTypeCode": [
    "KMP600I"
   ]
  },
  {
   "nodeCode": "TEST-1-1",
   "externalCode": "",
   "functionType": 1,
   "containerModelCode": [],
   "robotTypeCode": []
  }
 ],
 "code": "0",
 "message": "lang.wcs.common.ok",
 "success": true
}
```

-- Interface Response Description

- Success Response

```
{
  "code": "0",
  "message": "",
  "success": true,
  "data": ""
}
```

- Failure Response

```
{
  "code": "", //Error code
  "message": "", //Error description
  "success": false,
  "data": null
}
```

Common error description:

Error description	Error code	Message
Unexpected error	100001	UNKNOW ERROR
Invalid parameter	100002	INVALID PARAMETER
Required param missing	100404	MISS PARAM
Enum value not exists	100013	INVALID ENUM VALUE
No such node code	100204	NO SUCH NODE
Robot or robot type not exists	100301	ROBOT NOT EXISTS
Request id is duplicate	100408	API REQUEST ID IS DUPLICATE
Mission code is duplicate	100409	API MISSION CODE IS DUPLICATE